Operating Systems

(Sistemas Operativos)

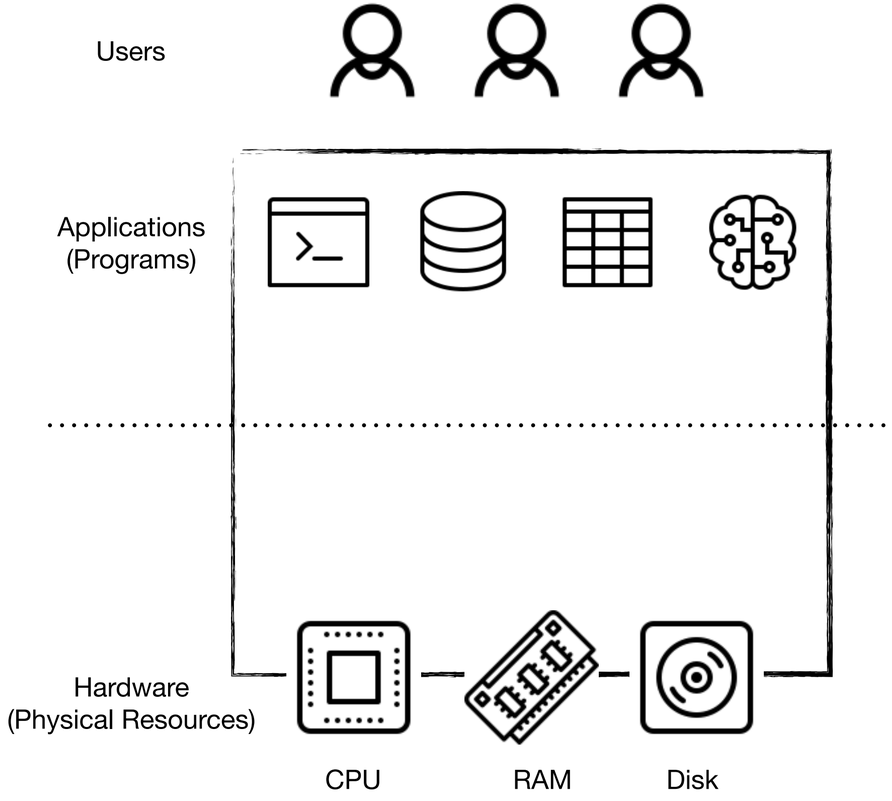
Introduction

University of Minho

2025-2026

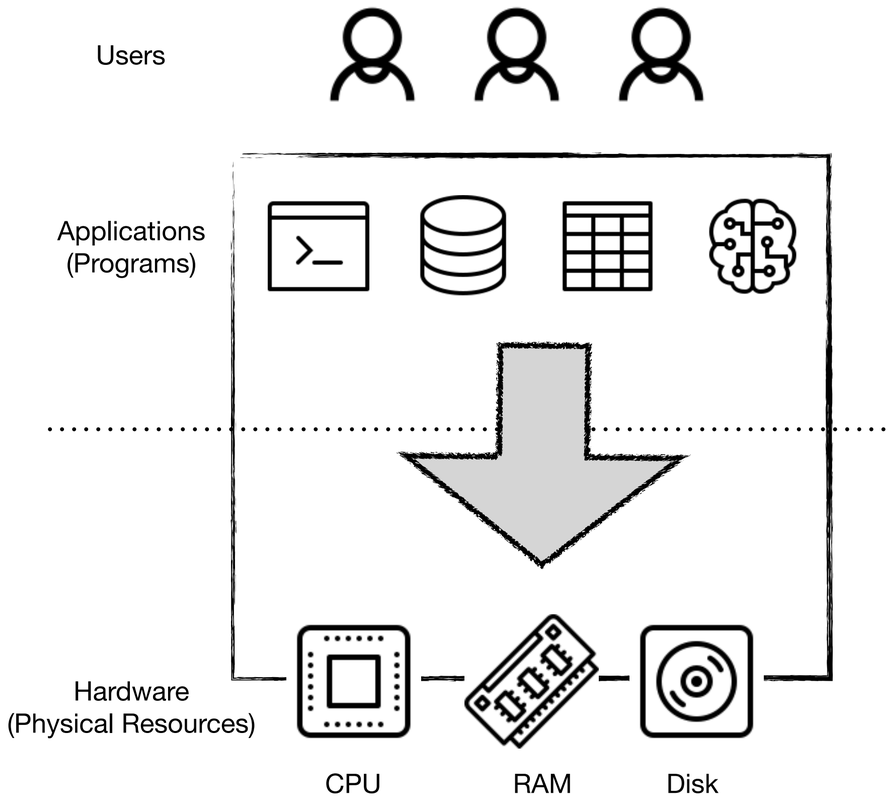
The Landscape

- Applications (Software)
  - Programs executing millions/billions of instructions per second (*e.g.*, text editors, databases, AI pipelines)
- Physical resources (Hardware)
  - CPU: where instructions are executed
  - Memory: stores volatile data structures
  - I/O Devices: external devices to interact with
    - Disk: to store data persistently
    - Keyboard, mouse, display, printer, …



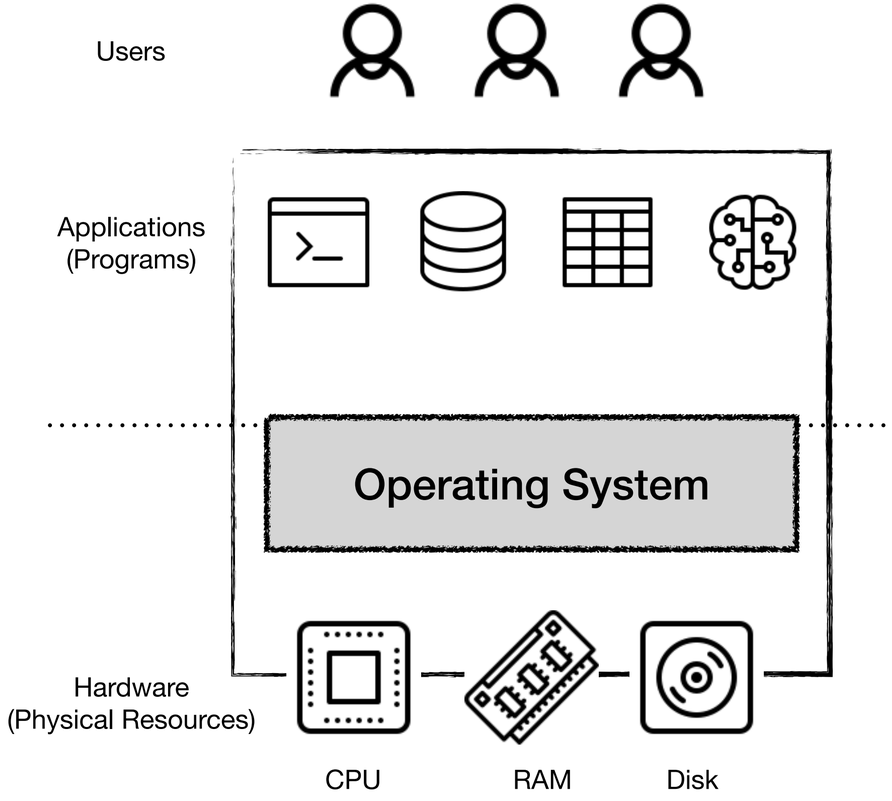
The Challenges

- Writing code that uses the hardware interfaces directly is challenging…
  - Different hardware devices and interfaces
  - Low-level languages (e.g., Assembly)
  - Time-consuming and error-prone
- and requires addressing several concerns
  - Efficient usage of hardware resources
  - Resource sharing (e.g., CPU, memory, disk) across multiple running programs
  - Security and isolation of running programs
  - …



Enters the Operating System (OS)

- Mediates the interaction between programs and hardware
- **Abstracts** physical resources into general, powerful, and easier-to-use virtual forms
- Offers **interfaces** (APIs) to use these resources (e.g., run programs, use memory and devices)
- **Manages** resources for multiple programs (e.g., concurrent programs, resource sharing)



Why Should I Do This Course?

- The OS is a powerful ally for efficiently running multiple programs on a computer. However:
- The incorrect use of OS tools and APIs may lead to
  - Inefficient programs (bad performance)
  - Incorrect programs (bugs, crashes, data corruption)
- To correctly use these tools and APIs, one must:
  - Try them - **practical classes**!
  - Understand them (means knowing the OS) - **theoretical classes**!
- And finally, because **Operating Systems are Fun**!
  - You will study algorithms, data structures, and optimizations applicable to different contexts (AI pipelines, databases, web services, apps, …)

Roadmap

Theory

- Introduction to Operating Systems
  - Design goal(s) and evolution of OSs
  - Modern OS stack, tools, and interfaces
- CPU Management
  - The process abstraction and API
  - CPU scheduling
- Memory Management
  - Address space management
  - Extending physical memory
- Persistence
  - I/O devices (HDDs, SSDs)
  - File Systems interface and design

Practice

- Persistence API
  - Creating, writing, and reading files
  - Improving programs' performance when accessing and persisting data
- Processes API
  - Creating processes
  - Sequential and concurrent programs
  - Executing programs
- Process Communication API
  - Efficiently exchanging data across
    - related processes
    - un-related processes

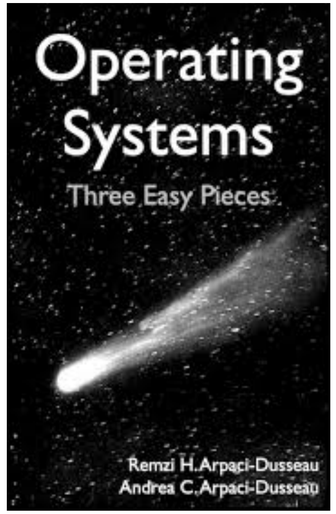
Assessment

- <u>Project (50%)</u>
  - Report: 10/05/2026
  - Presentation: 01/06/2026 - 05/06/2026

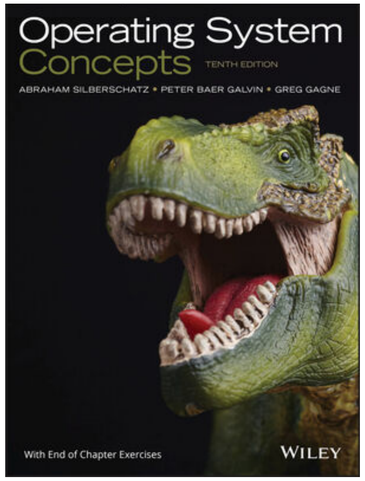
- <u>Written exam (50%) – minimum grade: 8 values</u>
  - 22/05/2025

Main References

- Remzi H. Arpaci-Dusseau, Andrea C. Arpaci-Dusseau.
  Operating Systems: Three Easy Pieces. Arpaci-Dusseau Books, 2018.



- Avi Silberschatz, Peter Baer Galvin, Greg Gagne.
  Operating System Concepts (10. ed). John Wiley & Sons, 2018.



Team (LEI + LCC)

- João Paulo (LEI+LCC) - jtpaulo@di.uminho.pt
- Ana Alonso (LEI + LCC) - ananunes@di.uminho.pt
- Rúben Daniel Adão (LEI + LCC) - d14301@di.uminho.pt
- Victor Fonte (LCC) - vff@di.uminho.pt
- Pedro Peixoto (LEI) - d14110@di.uminho.pt
- Tânia Esteves (LEI) - d12729@di.uminho.pt
- Nuno Faria (LEI) - d12956@di.uminho.pt
- Francisco Neves (LEI) - d8328@di.uminho.pt

Questions?